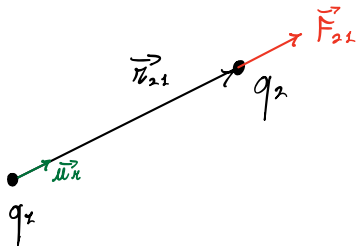
Legge di Coulomb



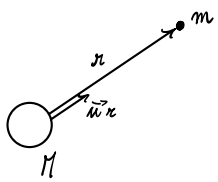
$\vec{r}_{21} = r\,\vec{u}_r$

$$\Rightarrow \boxed{\vec{F}_{21} = K\,\frac{q_1 q_2}{r^2}\,\vec{u}_r}$$ Legge di Coulomb

- $\vec{F}_{12} = -\vec{F}_{21}$ $\Rightarrow$ dirette come $\vec{u}_r$

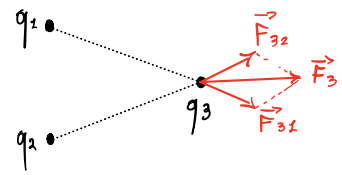
- intensità $e \xleftrightarrow{0.5\,\text{Å}} p$ $\Rightarrow F_{el} \cong 10^{-9}\,N$ $F_{gravitaz} \cong 10^{-47}\,N$

  ↘ è trascurabile nel caso delle cariche elementari



$$\vec{F}_{grav} = -\gamma\,\frac{m M}{r^2}\,\vec{u}_r$$

- sovrapposizione:

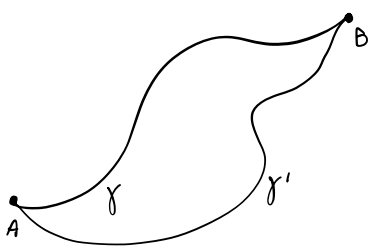


$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32} = \sum_{i=1}^{n} \vec{F}_{3i}$$

$\Rightarrow$ Principio di sovrapposizione degli effetti

- conservatività

$\vec{F}$ è centrale a simmetria sferica $\Rightarrow \vec{F}(\vec{r}) = F(r)\,\vec{u}_r$



Devo mostrare che $\mathcal{L}^{\gamma}_{AB} = -\Delta E_p(r)$

$$\mathcal{L}^{\gamma}_{AB} = \int_A^B \vec{F}(\vec{r})\,d\vec{l} = \int_A^B F(r)\,\vec{u}_r \cdot d\vec{l} = \int_A^B F(r)\,dr =$$

$$= \int_A^B \frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1 q_2}{r^2}\,dr = \frac{q_1 q_2}{4\pi\varepsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right) = E_p(r_A) - E_p(r_B)$$

$\Rightarrow$ la forza è conservativa!

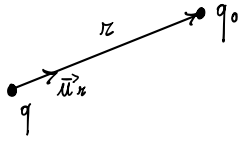
$$\Rightarrow \boxed{E_p(r) = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r} + C}$$

Energia potenziale elettrostatica (per la carica puntiforme)

($C = 0$ si pone di solito)

CAMPO ELETTROSTATICO

Consideriamo:



$$\boxed{\frac{\vec{F}(\vec{r})}{q_0} = \vec{E}(\vec{r})}$$

Campo elettrico

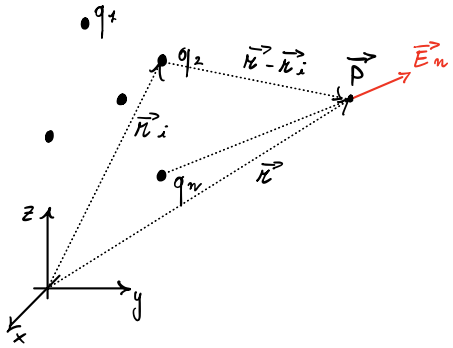
$$\Rightarrow \vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\vec{u}_r \quad \left[\frac{N}{C}\right]$$

Descrive le possibili interazioni nello spazio per via della carica q.

La carica q modifica il vuoto attorno a sé.

Possiamo anche vederlo come: $\vec{E}(\vec{r}) = \lim_{q_0 \to 0} \frac{\vec{F}(r)}{q_0}$

- Consideriamo



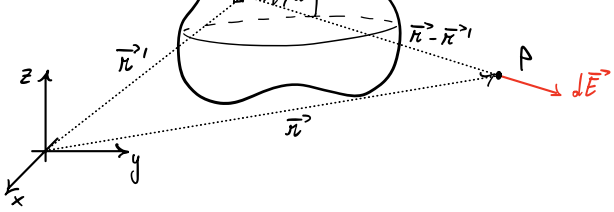
$$\boxed{\vec{E}_{tot} = \sum_{i=1}^{n} \vec{E}_i = \sum_{i=1}^{n} \frac{1}{4\pi\varepsilon_0}\frac{q_i}{|\vec{r} - \vec{r}_i|^3}(\vec{r} - \vec{r}_i)}$$

Campo prodotto da n-cariche

- Consideriamo



$$d\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{dq}{|\vec{r} - \vec{r}'|^2}(\vec{r} - \vec{r}')$$



$$\vec{E} = \int d\vec{E} = \int \frac{1}{4\pi\varepsilon_0} \frac{dq}{|\vec{r}-\vec{r}'|^3} (\vec{r}-\vec{r}')$$

$$\boxed{\vec{E} = \int_\tau \frac{1}{4\pi\varepsilon_0} \frac{\rho(\vec{r}')}{|\vec{r}-\vec{r}'|^3} (\vec{r}-\vec{r}')\, d\tau}$$

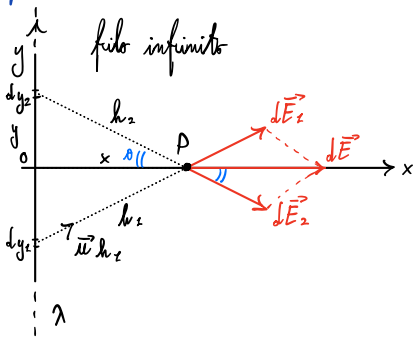
$$\rho(\vec{r}') = \frac{dq}{d\tau}$$

Campo di un volume continuo di carica

potrei fare lo stesso monodimensionalmente

e bidimensionalmente $\left(\sigma(\vec{r}') = \frac{dq}{dS}, \lambda(\vec{r}') = \frac{dq}{dl}\right)$

- Esempio



$$d\vec{E}_1 = \frac{1}{4\pi\varepsilon_0} \frac{\lambda\, dy_1}{h_1^2} \vec{u}_{h_1}$$

$$d\vec{E} = (dE_1 + dE_2)\cos\theta\, \vec{u}_x$$

$$dE = \frac{2}{4\pi\varepsilon_0} \frac{\lambda\cos\theta}{h^2} dy = \frac{1}{2\pi\varepsilon_0} \frac{\lambda\cos\theta}{\frac{x^2}{\cos^2\theta}} \frac{x}{\cos^2\theta} d\theta =$$

$$= \frac{1}{2\pi\varepsilon_0 x} \lambda\cos\theta\, d\theta$$

- $y = x\, tg\,\theta \longrightarrow dy = \frac{x}{\cos^2\theta} d\theta$
- $x = h\cdot\cos\theta \longrightarrow h = \frac{x}{\cos\theta}$

$\Big\}\ x = \text{cost}$

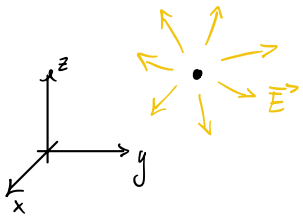
$$\Rightarrow E\,tot = \int dE = \int_0^{\frac{\pi}{2}} \frac{\lambda}{2\pi\varepsilon_0 x} \cos\theta\, d\theta = \frac{\lambda}{2\pi\varepsilon_0 x} \left[\sin\theta\right]_0^{\frac{\pi}{2}} = \frac{\lambda}{2\pi\varepsilon_0 x}$$

$$\boxed{\vec{E}(\vec{r}) = \frac{\lambda}{2\pi\varepsilon_0 r} \vec{u}_r}$$

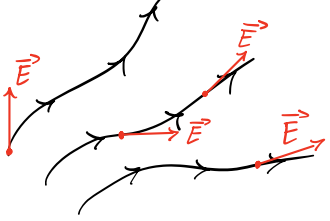
Campo elettrico di un filo infinito

055. $\vec{E}$ trovato $\propto \frac{1}{r}$

Linee di Campo

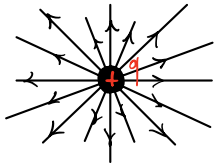


più comodo $\Rightarrow$



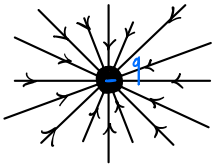
- tangenti a $\vec{E}$
- equiverse (stesso verso di $\vec{E}$)
- densità proporzionale a quella di $\vec{E}$
- non si intersecano

Sorgenti del campo $\vec{E}$

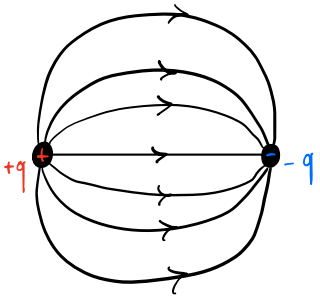


$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\vec{u}_r$$

Pozzi del campo $\vec{E}$



$$\vec{E}(\vec{r}) = -\frac{1}{4\pi\varepsilon_0}\frac{|q|}{r^2}\vec{u}_r$$



Dipolo elettrico

Consideriamo $|+q| = |-q| = q$

Osservato da lontano, le cariche è come se fossero unite, quindi un osservatore molto distante non dovrebbe percepire $\vec{E}$.

$\Rightarrow$ tutte le linee escono da $+q$ ed entrano in $-q$

Questo non vale se $|+q| \neq |-q|$

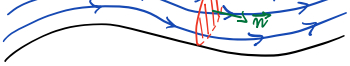
FLUSSO DI UN CAMPO VETTORIALE

Esempio del fluido



Il flusso è:

- $\propto dS$



- $\propto \vec{v}$
- max se $\vec{v} \parallel \vec{n}$
- nullo se $\vec{v} \perp \vec{n}$

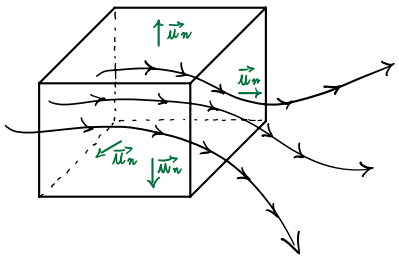
$$d\Phi = \vec{v} \cdot \vec{n}\, dS$$

$$\Phi_S(\vec{v}) = \int_S d\Phi = \int_S \vec{v} \cdot \vec{n}\, dS$$

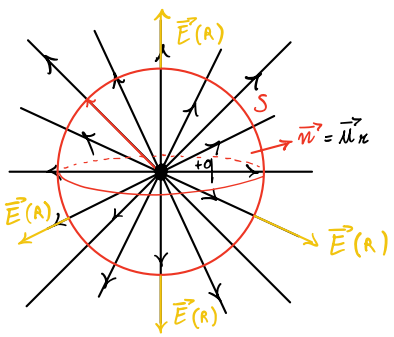
Flusso di un campo vettoriale

in caso di superficie chiusa:

$$\Phi_S(\vec{v}) = \oint_S \vec{v} \cdot \vec{n}\, dS$$



- Calcoliamo il flusso attraverso una sfera



$$d\Phi(\vec{E}) = \vec{E}(R) \cdot \vec{u}_n\, dS = \frac{1}{4\pi\varepsilon_0} \frac{q}{R^2} \underbrace{\vec{u}_n \cdot \vec{u}_n}_{=1} dS$$

$$\Phi(\vec{E}) = \int_S \frac{q}{4\pi\varepsilon_0 R^2} dS = \frac{q}{\cancel{4\pi\varepsilon_0 R^2}} \cdot \cancel{4\pi R^2} = \frac{q}{\varepsilon_0}$$